

G.3 Measuring persona drift with role PC1

In long, multi-turn conversations, we find that projecting activations on role PC1 also detects drift most often in conversations related to philosophy and therapy. In those conversation domains, the projection often trends towards lower values as the conversation continues or begins at a lower range to start.

On the other hand, we see that writing can occasionally begin with a lower projection but then increase, implying the model shifts back towards the Assistant. Coding conversations generally remain stable and do not show signs of persona drift.

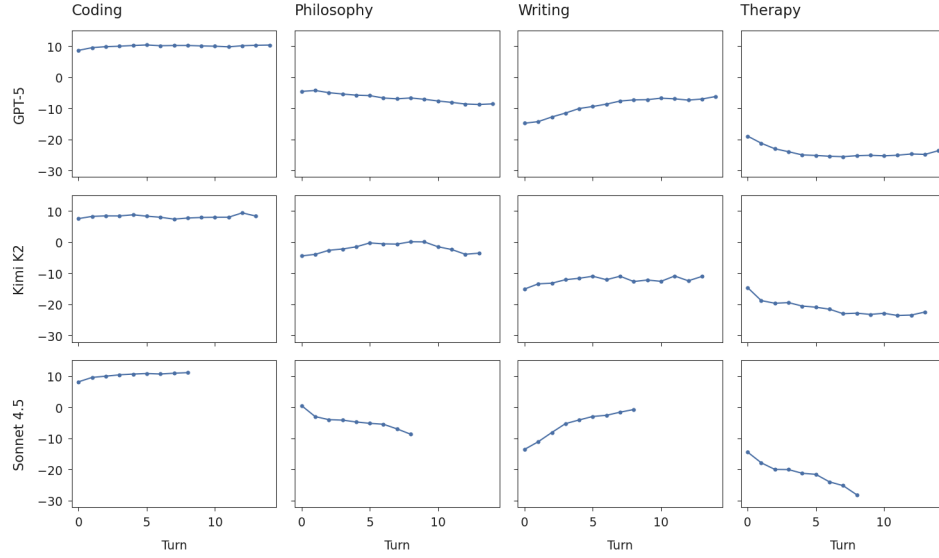


Figure 31: Projections on role PC1 for Gemma 2 27B in multi-turn conversations across domains, with responses averaged per turn.

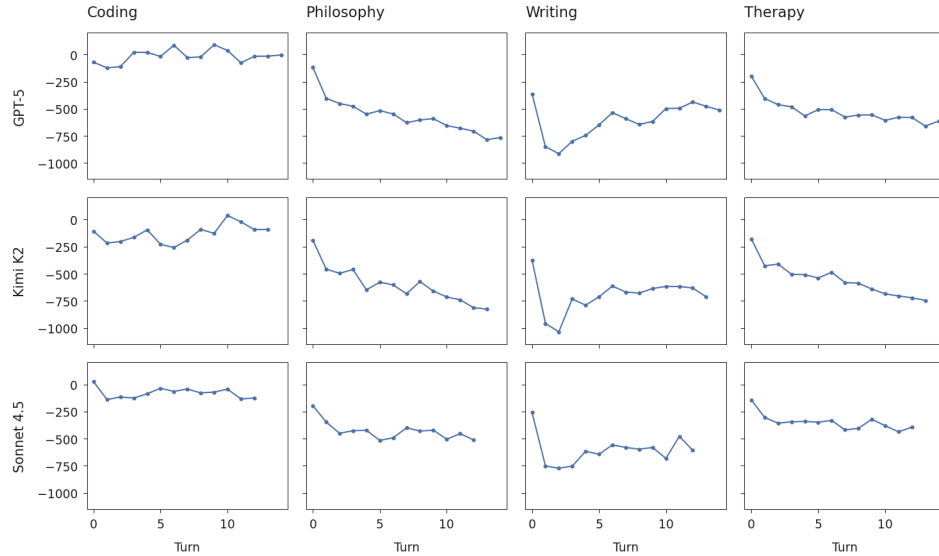


Figure 32: Projections on role PC1 for Qwen 3 32B in multi-turn conversations across domains, with responses averaged per turn.

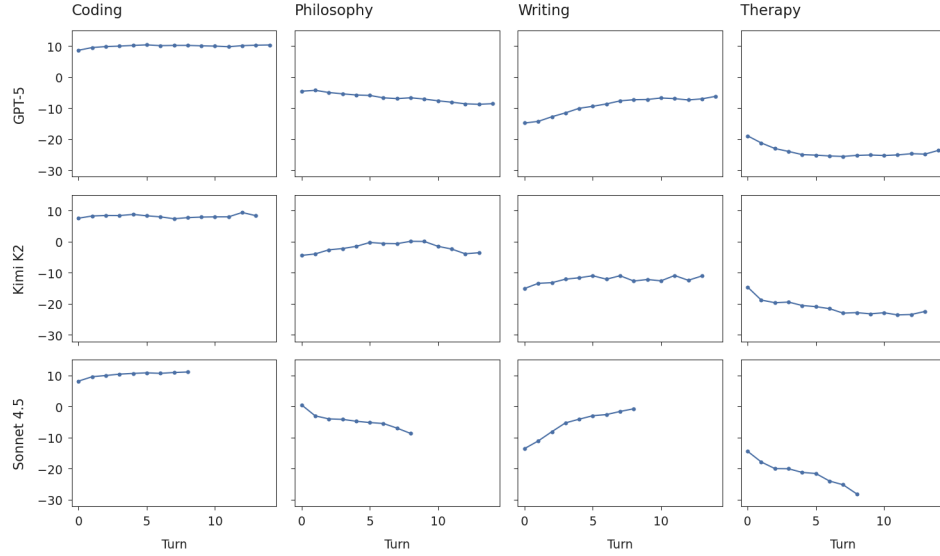


Figure 33: Projections on role PC1 for Llama 3.3 70B in multi-turn conversations across domains, with responses averaged per turn.

G.4 Activation capping along role PC1

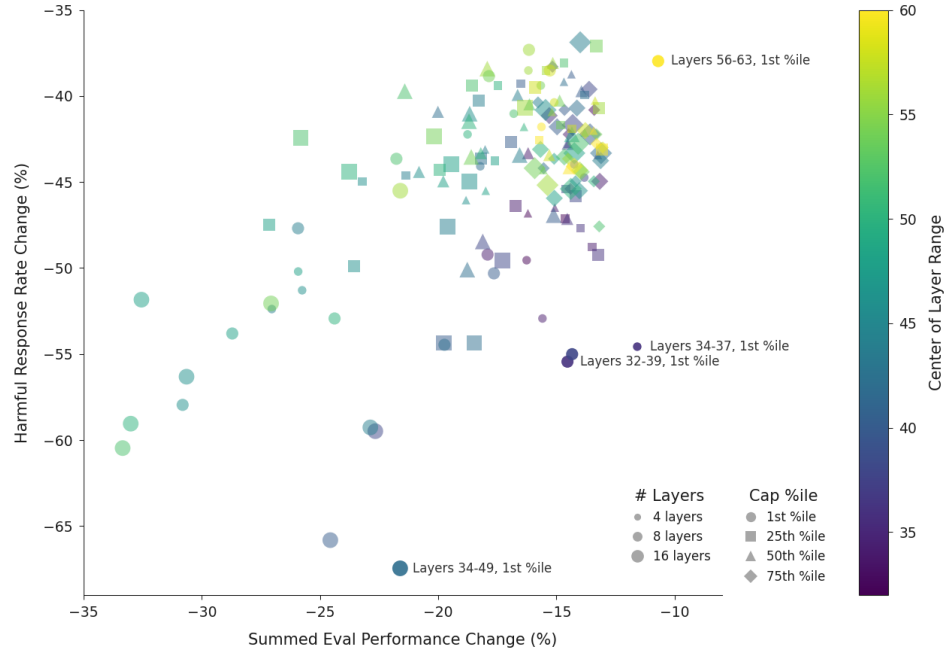


Figure 34: Changes in harmful response rates (on persona-based jailbreak prompts) and capabilities eval performance (averaged over a suite of evaluations—IFEval, MMLU Pro, GSM8k, EQ-Bench) for different activation capping settings along role PC1, varying the layer range and cap threshold (given in terms of a percentile, relative to the activations from our dataset used to compute role vectors), for Qwen 3 32B.

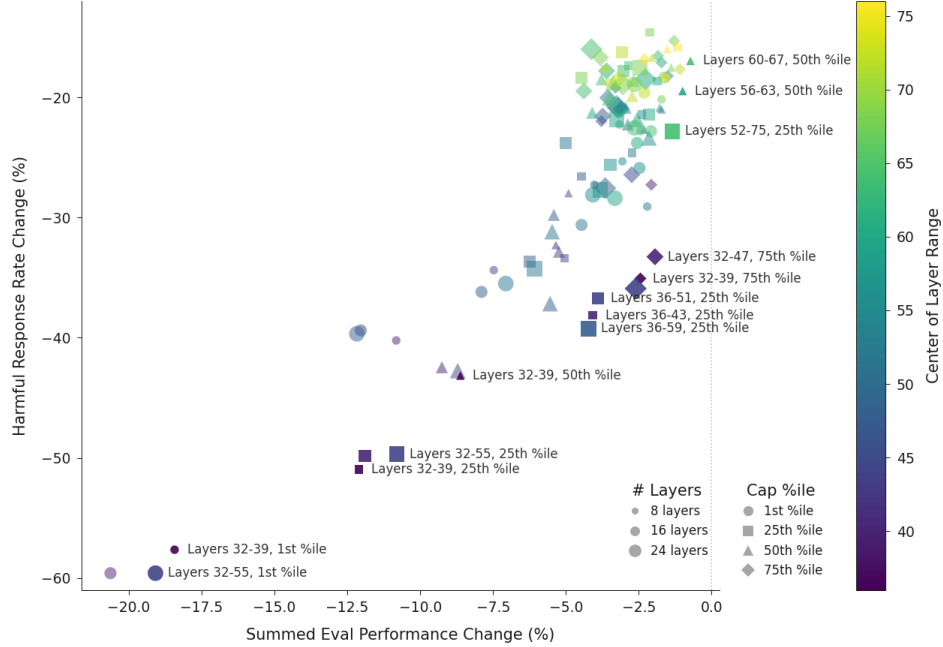


Figure 35: Changes in harmful response rates (on persona-based jailbreak prompts) and capabilities eval performance (averaged over a suite of evaluations—IFEval, MMLU Pro, GSM8k, EQ-Bench) for different activation capping settings along role PC1, varying the layer range and cap threshold (given in terms of a percentile, relative to the activations from our dataset used to compute role vectors), for Llama 3.3 70B.

In our activation capping experiments, we also tried capping along role PC1 in Qwen 3 32B and Llama 3.3 70B. We calibrated our activation caps the same way, by using percentiles from the distribution of projections of activations from our role/trait rollouts, and swept the same layer ranges and widths.

We note that for this experiment, we used VLLM to run our steering experiments over the TRANSFORMERS library for speed. However, when comparing runs across the two implementations, we observed consistently 2-3% worse performance on the various evaluations using VLLM. Hence, in these results both models show a larger performance hit across capabilities with activation capping (particularly in Qwen). We think that performance would likely improve using TRANSFORMERS instead. We present the results here to discuss the effective steering settings along the Pareto frontier in relative terms.

In Qwen, we find that stricter caps seem to be effective when using role PC1 to activation cap (the 1st percentile rather than the 25th), at a slightly earlier layer range than with the Assistant Axis. However, using role PC1 is less effective at reducing the rate of harmful responses to jailbreaks.

In Llama, we similarly find that role PC1 is less effective at reducing the rate of harmful responses to jailbreaks. Most of the steering settings on the Pareto frontier involve earlier layer ranges than using the Assistant Axis, but also mainly consist of activation caps calibrated to the 25th percentile of projections.

G.5 Discussion

Overall, these results suggest that role PC1 and the Assistant Axis have high cosine similarity across layers, similar steering effects, and ability to detect persona drift. However, there are still differences in performance, notably in how activation capping role PC1 is worse at mitigating persona-based jailbreaks. Since it is not guaranteed that role PC1 would capture the same meaning in every model, we suggest using the contrast vector method to obtain the Assistant Axis in future implementations.